

• Einstein - Hilbert action:

$$S_{EH} = \int d^d x \sqrt{-g} \frac{1}{2} M_{Pl}^2 R$$

$\sim$ Ricci scalar
 $M_{Pl} = \frac{1}{\sqrt{8\pi G}}$

$$R = 2\Gamma - 2\Gamma + \Gamma\Gamma - \Gamma\Gamma$$

$$R = g^{\mu\nu} R_{\mu\nu} = R^\alpha{}_{\mu\alpha\nu}$$

$\rightarrow$ add matter $\Rightarrow S = \int d^d x \sqrt{-g} \left\{ \frac{1}{2} M_{Pl}^2 R + \mathcal{L}_M \right\}$

$\underbrace{\int d^d x \sqrt{-g}}_{\text{diffeo}^m \text{ inv. measure}}$

$$\Rightarrow \frac{\delta S}{\delta g_{\mu\nu}} : \frac{\delta}{\delta g_{\mu\nu}} \left(\sqrt{-g} g^{\mu\nu} R_{\mu\nu} \right) = -\sqrt{-g} \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right)$$

$\underbrace{\quad}_{\text{Einstein tensor}}$

$\rightarrow \Lambda, R$: leading terms in a derivative expansion
 $\mathcal{E}$ - $\mathcal{H}$ action $\Rightarrow$ low-energy effective theory for gravity

$\rightarrow$ 'Theorem': Einstein GR is a unique low energy interacting theory for a massless spin-2 particle (graviton = gravitational waves)

• L E F T

$\rightarrow$ write down every local operator consistent with symmetries

$\rightarrow$ components: $g^{\mu\nu}$ (field content), symmetry diffeomorphism,
local operators: $\Lambda = \text{L E F T energy cutoff}$

$$\Rightarrow \mathcal{L}_g = \mathcal{F}[R^\mu{}_{\nu\rho\sigma}, \nabla_\mu] \quad \left[S = \int d^d x \sqrt{-g} (\mathcal{L}_g + \mathcal{L}_M) \right]$$

$\rightarrow$ no independent scalar can be built out of the metric
via diffeo^m invariance (EEP)

→ set scale to be $\Lambda^4 = \mathcal{L}$

$$\Rightarrow \mathcal{L}_g = \Lambda^d \left[\Lambda^4 + \frac{R}{\Lambda^2} + \frac{a R^2}{\Lambda^4} + b \frac{R_{\mu\sigma} R^{\mu\sigma}}{\Lambda^4} + c \frac{R_{\mu\nu\sigma\rho} R^{\mu\nu\sigma\rho}}{\Lambda^4} \right]$$

cosmological const.

unique leading corrections to GR

• Fields: $\mathcal{M}$: $S = \int -\frac{1}{2} \frac{1}{g_{\mu\nu}} \text{tr} (F_{\mu\nu} F^{\mu\nu})$

$$\rightarrow F_{\mu\nu}^a = \frac{1}{g} \left(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c \right), \quad A_\mu \rightarrow \frac{1}{g} A_\mu$$

$\frac{1}{g}$ absorbed into A_μ

→ small g : weak coupling $\sim \hbar$

⇒ add a new weak coupling to $\mathcal{L}_g$:

$$\mathcal{L}_g = \frac{\Lambda^d}{g^2} \left[\Lambda^4 + \frac{R}{\Lambda^2} + \frac{a R^2}{\Lambda^4} + b \frac{R_{\mu\sigma} R^{\mu\sigma}}{\Lambda^4} + c \frac{R_{\mu\nu\sigma\rho} R^{\mu\nu\sigma\rho}}{\Lambda^4} \right]$$

• String theory: $\Lambda = M_s = \frac{1}{\sqrt{\alpha}} \longleftrightarrow g_s$

→ low energy scattering amplitude

$$\mathcal{A}\left(\frac{E}{\Lambda}\right) = \mathcal{A}_0(E) \left(1 + \# \frac{E^2}{\Lambda^2} + \# \frac{E^4}{\Lambda^4} \right)$$

$$\rightarrow S_{\text{string}}^{\text{10d}} = \int d^{10}x \left[\frac{\Lambda^8}{g^2} \left\{ R + \alpha' R^2 + \dots + \alpha'^2 R^3 + \dots + \alpha'^3 R^4 + \dots \right\} \right]$$

counts loops M_{Pl}

$M_{Pl}^8 = \frac{M_s^8}{g_s^2}$

$g_s \ll 1, M_{Pl} \gg M_s$

→ old-fashioned case:

$$S = \int d^d x \frac{1}{2} M_{Pl}^{d-2} \left\{ R + a \frac{R^2}{M_{Pl}^2} + \dots + b \frac{R^3}{M_{Pl}^3} + \dots \right\}$$

• Example: loops for matter

$$S_\phi = \int d^4x \sqrt{-g} \left\{ -\frac{1}{2} M_{Pl}^2 R - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} M \phi^2 \right\}$$

$\Rightarrow$ to quantise :

we are in curved space

$$\mathcal{Z} = \int \mathcal{D}\phi e^{iS_\phi} = \exp \left(-i \int d^4x \sqrt{-g} \left[\frac{1}{2} M_{Pl}^2 R + \frac{1}{2} \phi \ln [\square + M^2] \right] \right)$$

$\rightarrow$ cutoff: Λ_z

$$\Rightarrow \int d^4x \left\{ \Lambda_z^4 + \Lambda_z^2 R + \ln \Lambda_z R_{\mu\nu}^2 + M^2 R + \ln(M) R_{\mu\nu}^2 \right\}$$

$\Rightarrow$ add a term to S_ϕ

$$S_\phi = \int d^4x \sqrt{-g} \left\{ -\frac{1}{2} M_{Pl}^2 R - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} M \phi^2 + c R_{\mu\nu}^2 \right\}$$

• Quantising GR

$\rightarrow$ recall: Maxwell $\sim$ covariant $\sim \partial^\mu A_\mu = 0$

add a gauge fixing term:

$$S_{g-f} = \int d^d x \left[\frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right] \quad \text{Feynman gauge } \xi = 1$$

$$\Rightarrow \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} (\partial_\mu A^\mu)^2$$

$$= -\frac{1}{2} \partial_\mu A_\nu (\partial^\mu A^\nu - \partial^\nu A^\mu) - \frac{1}{2} (\partial_\mu A^\mu)^2$$

$$= -\frac{1}{2} \partial_\mu A_\nu \partial^\mu A^\nu$$

$$= \frac{1}{2} A_\mu \square A^\mu$$

Gravitons propagating in Minkowski

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2}{M_{Pl}} h_{\mu\nu}$$

$$S = \int d^d x \frac{1}{2} M_{Pl}^2 \sqrt{-g} \mathcal{R}$$

$$= \int d^4 x \left\{ -\frac{1}{2} \partial_\alpha h_{\mu\nu} \partial^\alpha h^{\mu\nu} + \frac{1}{2} \partial_\mu h \partial^\mu h - \partial_\mu h \partial_\nu h^{\mu\nu} + \partial_\alpha h_{\mu\beta} \partial^\beta h^{\mu\alpha} \right\}$$

$h = h^\mu_\mu$

→ gauge choice : de Donder (harmonic gauge)

$$\partial^\mu \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) = 0$$

$$\rightarrow x^\mu \rightarrow x^\mu + \varepsilon^\mu, \quad h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \varepsilon_\nu + \partial_\nu \varepsilon_\mu$$

$$\Rightarrow \partial^\mu \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) \rightarrow \partial^\mu \left(h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h \right) + \square \varepsilon_\mu$$

$$\Rightarrow S = S_{GR}^{free} + \int d^d x \left\{ \underbrace{-\frac{1}{2} \left(\partial^\mu h_{\mu\nu} - \frac{1}{2} \partial_\nu h \right)^2}_{\phi \square \phi} \right\} + S_{\mathcal{M}}$$

$$= \int d^d x \frac{1}{2} h_{\mu\nu} \square \left(h^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} h \right) + \frac{h_{\mu\nu} T^{\mu\nu}}{M_{Pl}} + \phi \mathcal{J}$$

+ non-linear GR interaction

$$\Rightarrow \langle 0 | \hat{T} \exp \left\{ i \int d^d x \frac{h_{\mu\nu} T^{\mu\nu}}{M_{Pl}} \right\} | 0 \rangle$$

$$= \int \mathcal{D} h_{\mu\nu} \exp \left\{ i \int d^d x \frac{1}{2} h_{\mu\nu} \square \left(-h^{\mu\nu} - \frac{1}{2} \eta^{\mu\nu} h \right) + \frac{h_{\mu\nu} T^{\mu\nu}}{M_{Pl}} \right\}$$

$$\Rightarrow Z[J] = e^{iW[J]} = \langle 0 | T \exp \left\{ i \int d^d x J(x) \phi(x) \right\} | 0 \rangle$$

$$\square h_{\mu\nu} = -\frac{1}{M_{Pl}^2} \left(\tau^{\mu\nu} - \frac{1}{(d-2)} \eta^{\mu\nu} \tau \right)$$

$$\Rightarrow h_{\mu\nu} = -\frac{1}{M_{Pl}^2} \frac{1}{\square + i\epsilon} \left(\tau^{\mu\nu} - \frac{1}{(d-2)} \eta^{\mu\nu} \tau \right)$$

$$\left[\square h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} \square h = -\frac{\tau^{\mu\nu}}{M_{Pl}^2}, \quad \left(1 - \frac{d}{2}\right) \square h = -\frac{\tau}{M_{Pl}^2} \right]$$

$$\rightarrow \text{non-relativistic source} \quad \tau^{00} \neq 0, \quad \tau^{0i} = 0 = \tau^{ij}$$

$$W[\tau_{\mu\nu}] = \int d^d x \left(-\frac{1}{2M_{Pl}^2} \tau^{\mu\nu} \frac{1}{\square + i\epsilon} \tau^{\mu\nu} + \frac{1}{d-2} \tau \frac{1}{\square + i\epsilon} \tau \right)$$

$$\equiv W[J] = \int d^d x \left(-J \frac{1}{\square + i\epsilon} J \right)$$

Curved background (background field method)

$$\rightarrow g_{\mu\nu} = \underbrace{\bar{g}_{\mu\nu}}_{\text{keep background diffeomorphisms}} + \underbrace{2 \frac{h_{\mu\nu}}{M_{Pl}^2}}_{\text{break with g-f terms}}$$

covariant derivative associated with $\bar{g}_{\mu\nu}$

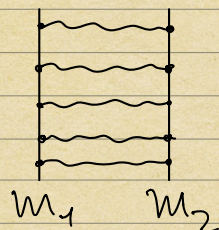
$$\rightarrow \text{gauge choice} : \tau^\mu = \bar{\nabla}^\nu \left(h_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} h \right) \quad h = \bar{g}^{\mu\nu} h_{\mu\nu}$$

$$\Rightarrow S = S_{GF} - \frac{1}{2} \int d^d x \sqrt{-g} g^{\mu\nu} \left(\bar{\nabla}^\alpha \left(h_{\mu\alpha} - \frac{1}{2} \bar{g}_{\mu\alpha} h \right) \right) \left(\bar{\nabla}^\beta \left(h_{\nu\beta} - \frac{1}{2} \bar{g}_{\nu\beta} h \right) \right)$$

$$\rightarrow Z = \int \mathcal{D}g_{\mu\nu} \delta(\mathcal{F}^\mu) \det \left[\frac{\delta \mathcal{F}^\mu}{\delta \varepsilon^\nu} \right] e^{i\int \mathcal{L}_R} \quad \text{Zadev - Popov}$$

$$= \int \mathcal{D}g_{\mu\nu} \int \mathcal{D}\lambda_\mu \exp \left\{ i \int d^d x \lambda_\mu \mathcal{F}^\mu \right\} \\ \times \exp \left\{ i \int d^d x \sqrt{-g} \left[\bar{\varepsilon}^\mu \square \varepsilon_\mu - \bar{\varepsilon}^\mu \varepsilon^\nu \bar{R}_{\mu\nu} \right] \right\}$$

• Example: 2 masses:



tadpole diagram

graviton exchanged
multiple possibilities

$$\mathcal{L} = \int d^4 x \sqrt{-g} \left\{ \frac{1}{2} M_{Pl}^2 R - \frac{1}{2} (\nabla_\mu \phi)^2 - \frac{1}{2} M^2 \phi^2 \right\} \quad g_{\mu\nu} = \eta_{\mu\nu} + 2 \frac{h_{\mu\nu}}{M_{Pl}^2}$$

$$\rightarrow \text{tadpole} \quad \frac{h_{\mu\nu} T^{\mu\nu}(\phi)}{M_{Pl}^2} \quad \frac{1}{M_{Pl}^2} \quad \text{[diagram of a tadpole on a wavy line]}$$

$$\rightarrow \frac{1}{M_{Pl}^2} h^2 T \quad \text{[diagram of a tadpole on a wavy line]} \quad \frac{1}{M_{Pl}^2} \quad \text{[diagram of a tadpole on a wavy line]}$$

$$\rightarrow \text{[diagram of two vertical lines m1 and m2 connected by a wavy line]} \Rightarrow \text{[diagram of two vertices connected by a wavy line]}$$

$\frac{\delta^2 \Gamma_{1-PI}}{\delta g_{\mu\nu} \delta g_{\alpha\beta}}$
 $\frac{\delta^3 T_{1-PI}}{\delta g_{\mu\nu} \delta \phi \delta \phi}$

$$\text{[diagram of a wavy line]} = \text{[diagram of a wavy line with a circle]} + \text{[diagram of a wavy line with a shaded circle]} + \text{[diagram of a wavy line with a dashed circle]} + \text{[diagram of a wavy line with a shaded dashed circle]}$$

$$\text{[diagram of a vertex on a wavy line]} = \text{[diagram of a vertex on a wavy line with a shaded circle]} + \text{[diagram of a vertex on a wavy line with a shaded dashed circle]}$$

$\rightarrow$ one-loop $\begin{array}{c} \text{---} \text{---} \text{---} \\ \mu\nu \quad \quad \alpha\beta \end{array} \sim \frac{1}{32\pi^2 \mu_{Pl}^2} q^4 \left[\frac{21}{120} (\eta_{\mu\alpha} \eta_{\nu\beta} + \eta_{\mu\beta} \eta_{\nu\alpha}) \right.$
 $\left. \langle T h_{\mu\nu} h_{\alpha\beta} \rangle + \frac{1}{120} \eta_{\mu\nu} \eta_{\alpha\beta} \right] \times \ln\left(\frac{q^2}{\mu^2}\right)$